

Quiz 2

ECON 320 – Introduction to Mathematical Economics

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Instructions

- Answer all questions. Show all steps clearly to receive full credit.
 - You may use diagrams where appropriate.
 - Total time: 20 minutes.
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1. Consider the system:

$$\begin{cases} x^2 + y + z = \alpha \\ xy - z = \beta \\ y^2 + z^2 = 1 \end{cases}$$

- (a) Write the Jacobian matrix of the system with respect to (x, y, z) .
- (b) State the condition under which the Implicit Function Theorem guarantees the existence of differentiable functions $(x(\alpha, \beta), y(\alpha, \beta), z(\alpha, \beta))$.
- (c) Derive an expression for $\frac{\partial(x, y, z)}{\partial \alpha}$ using Cramer's rule.

2. Suppose the transformation is defined by:

$$u = x^2 + y^2 + z^2, \quad v = xy + yz + zx, \quad w = xyz.$$

- (a) Construct the Jacobian $\frac{\partial(u, v, w)}{\partial(x, y, z)}$.
- (b) Show that the determinant vanishes when at least two of (x, y, z) are equal.
- (c) Explain the implication for whether the transformation is locally invertible.

3. Let the system be:

$$\begin{cases} F_1(x, y, \theta) = x^2 + y^2 - \theta = 0 \\ F_2(x, y, \theta) = xy - 1 = 0 \end{cases}$$

- (a) Verify that the Jacobian determinant with respect to (x, y) is nonzero at $(x, y) = (\sqrt{2}, 1/\sqrt{2}), \theta = 3$.
- (b) Use the Implicit Function Theorem to argue for local differentiability of $(x(\theta), y(\theta))$.
- (c) Compute $\frac{dx}{d\theta}, \frac{dy}{d\theta}$ at this point.